What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Hemostasis

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Learning Objectives (5)

After completing this brick, you will be able to:

1. List the steps involved in primary (platelet plug) and secondary (fibrin clot) hemostasis.
2. Compare and contrast the extrinsic, intrinsic, and common pathways (clotting or coagulation cascade).
3. Describe the importance of Vitamin K and calcium in coagulation.
4. Describe the process and regulation of fibrinolysis and clot resolution.
5. Discuss common disorders affecting hemostasis.

CASE CONNECTION

GG is a 19-year-old female with symptomatic gallstones who is being evaluated in your preceptor's general surgery clinic. You are asked to help with her history and physical exam to help arrange her laparoscopic cholecystectomy.

Knowing that a thorough medical history is the most reliable screening tool for identifying potential coagulation disorders, you ask a series of questions. She describes having heavy menses, soaking through pads every 1-2 hours. She also has frequent nosebleeds and frequent bruising. Her gums bleed each time she brushes her teeth. She thought these symptoms were normal because her mother has the same symptoms.

The team orders a CBC, revealing a normal platelet count. PT and aPTT labs are also normal.

Could she still have a bleeding disorder?

GO TO CONCLUSION ↓

What Is Hemostasis?

The mechanism by which bleeding comes to an end is called hemostasis (*heme* =blood, *stasis* = standing still), also known as "clotting." It requires the precise and collective [illegible] (the coagulation cascade), the vascular wall, and platelets. Equally as important as making a clot (thrombus) is the ability to stop the formation of clots or else break them down through fibrinolysis. Otherwise, we would become a huge clot right after a papercut!

When the hemostatic process is working, clotting happens normally. But when there is a disturbance, we see the potential for two kinds of problems:

- Patients with **overactive clotting mechanisms or deficient anticlotting systems** are at higher risk for thrombosis, which is the pathological formation of clots in blood vessels (thrombophilia/hypercoagulability). This can result in strokes, deep vein thrombosis, or heart attacks.
- Conversely, patients with **deficient clotting mechanisms** have an increased tendency to bleed (hypocoagulability/insufficient clotting). Examples are hemophilia, von Willebrand disease, and inherited thrombocytopenia (low platelet count).

What kind of dysfunctions might predispose a patient toward thrombosis?

Hemostasis overview (LO #1)

When a blood vessel is damaged, a clot forms via vessel constriction, platelet plug formation, and fibrin formation. Primary hemostasis involves platelets and von Willebrand factor, a crucial blood protein, which form a platelet plug. Secondary hemostasis involves the formation of a fibrin clot through the activation of coagulation factors.

Primary hemostasis (temporary platelet plug)

Step 1: Blood Vessel Constriction

Immediate constriction of the blood vessel to minimize blood loss. Almost immediately after a vessel is damaged, regional small arterioles constrict, driven by release of the cytokine **endothelin** and inflammatory markers from endothelial cells (Figure 1).

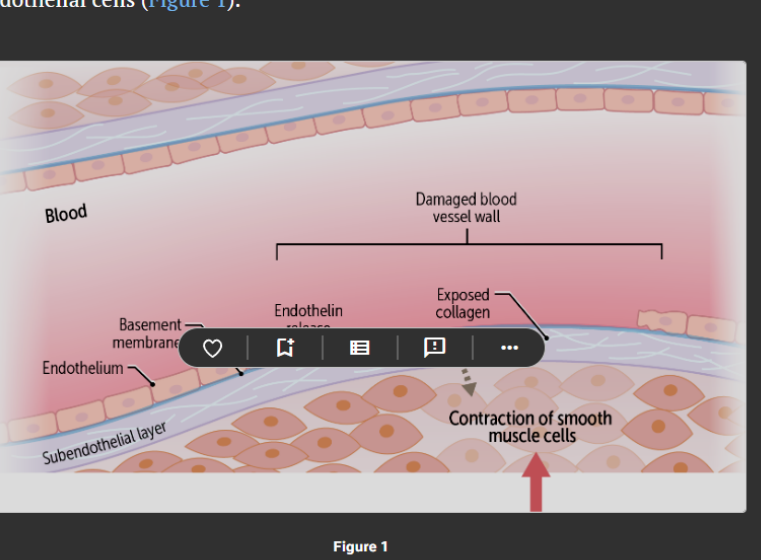


Figure 1

This constriction reduces the amount of blood loss at the site of injury, making it easier for the coagulation factors and platelets to come in direct contact with each other and with the injured endothelium so that the processes of platelet plug formation and fibrin formation can begin.

Step 2: Platelet Adhesion

Platelets stick to exposed collagen and subendothelial matrix. This process, called **adhesion**, involves von Willebrand factor (vWF). VWF, secreted by endothelial cells, acts as a "molecular glue" at the damage site binding to both exposed collagen and activated platelet glycoproteins receptors (Glycoprotein Ib).

Step 3: Platelet Activation

Additional platelets are recruited to form a platelet plug. Binding of vWF to Glycoprotein Ib activates the platelet, leading to production of thromboxane A2, an important mediator in platelet aggregation.

Thromboxane A2:

1. causes platelet granule release. The released granules contain substances such as ADP, calc[illegible] others.
2. activates glycoprotein IIb/IIIa, the fibrinogen receptor on the platelet surface
3. changes the configuration of the platelet phospholipid membrane to make it more conducive to coagulation.
4. is released into the plasma where it activates and recruits other platelets, promoting platelet aggregation.
5. causes vasoconstriction.

Step 4: Platelet aggregation

The activated platelets bind to fibrinogen via Glycoprotein IIa/IIIb receptors forming a loose clot. The result is a **soft platelet plug** that covers the hole in the damaged vessel to stop bleeding (Figure 2). Although this is a great way to achieve temporary hemostasis, the platelet plug is not stable and it quickly falls apart.

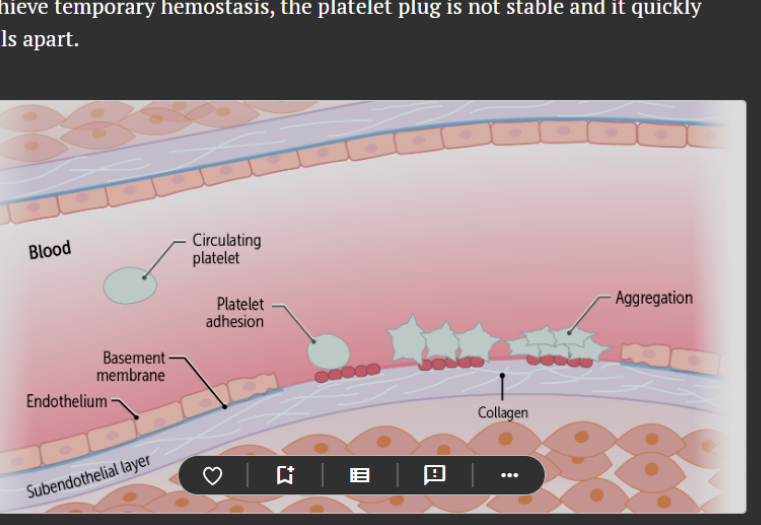


Figure 2

What are the steps in primary hemostasis?

Secondary Hemostasis (crosslinked fibrin clot formation) (LO #2)

Because the initial clotting from primary hemostasis only results in temporary clots, an additional process is needed to stabilize the clot— secondary hemostasis

The main goal is stabilization of the initial platelet plug to a fibrin clot. This involves sealing the platelet plug in place with a glue-like molecule called fibrin. Fibrin is formed [illegible] **coagulation cascade** (Figure 3).

Secondary hemostasis and the coagulation cascade are the same thing. We will first cover the steps in general and then discuss the coagulation cascade in detail later in this brick.

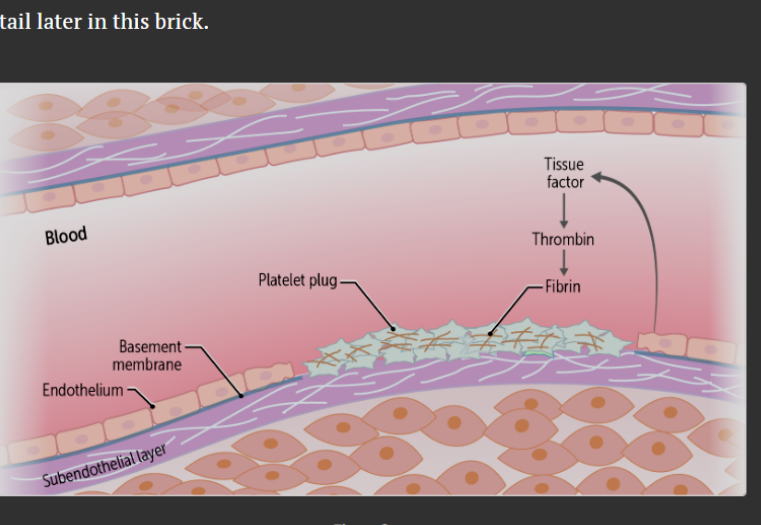


Figure 3

What is [illegible] ostasis?

Steps of the Coagulation Cascade (Secondary hemostasis)

The coagulation cascade forms a stable clot at the injury site by producing fibrin, which solidifies the platelet plug. It is a chain of enzymes each triggering the next step in a "cascade". (activated proteins are designated by the letter "a" following the Roman numeral) (Figure 4).

The cascade involves multiple enzymes, or factors present in the blood. Figure 4 shows them arranged in a Y-shaped pattern. Clotting involves the intrinsic arm, activated by exposure of clotting factors to the negative charges of collagen (left), and the extrinsic arm, activated by substances released by tissue injury (right). Both pathways converge at the final common pathway in the middle. While it appears that either intrinsic or extrinsic pathways might operate independently to make fibrin, in reality, both pathways operate simultaneously to form long term stable clots.

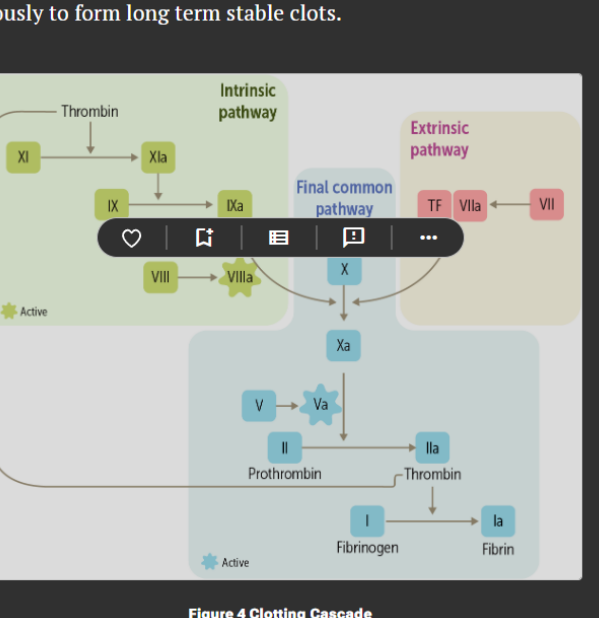


Figure 4 Clotting Cascade

Extrinsic Pathway (factors extrinsic to – outside– the blood)

- Triggered by tissue factor (TF) released from damaged tissues.
- Involves Factor VII (Factor 7) and TF (Factor 3- which is not normally in the blood).
- Rapid response.

The cascade is triggered by a substance called **tissue factor** (TF, Factor III), which sits at the top of the **extrinsic pathway** in Figure 4. Normally, tissue factor is hidden from the bloodstream until the cells of a vessel are damaged and tissue factor is released. Once released by endothelial cells (and some other places), it initiates the coagulation cascade along the extrinsic arm. The process then proceeds through the final common pathway, starting at Factor X, ultimately converting fibrinogen into fibrin.

Physiologically, the t[illegible] itiation phase of coagulation: it only generates a small amount of thrombin, but it is this **thrombin that loops back to the intrinsic pathway** to activate other clotting factors and initiate the amplification phase.

Intrinsic pathway (factors present in blood)

- Activated by trauma to the endothelial wall, which exposes collagen, a negatively charged surfaces.
- Involves Factors XII, XI, IX, and VIII.
- Amplified by thrombin
- Slower but more complex.

The extrinsic pathway generates enough fibrin to form a clot. However to amplify the process, thrombin from the final common pathway loops back to initiate the intrinsic pathway, increasing fibrin production. This boosts the coagulation process through the left side of the pathway, ensuring enough fibrin is produced to stabilize or "glue" the platelet plug together.

The intrinsic pathway can be triggered by exposure to negatively charged surfaces like bacterial infiltration, phospholipids, or circulating lipoproteins, the latter explaining why hyperlipidemia contributes to atherosclerosis.

CLINICAL CORRELATION

Factor XII (Hageman factor) is of interest. Although Factor XII deficiency prolongs clotting time in vitro, it does not cause clinical bleeding issues, as it is not involved in physiologic coagulation. This is the reason why it is not shown in Figure 4.

Common Pathway

- Convergence point of extrinsic and intrinsic pathways.
- Involves Factors X, V, II (prothrombin), and I (fibrinogen).
- Leads to formation of a stable fibrin clot.

The goal of this step is to produce fibrin, forming a crosslinked fibrin clot. The common pathway, activated by either intrinsic or extrinsic pathways, converts factor X to Xa. This leads to thrombin (factor II) production, which converts fibrinogen (Factor 1) to fibrin. Fibrin then glues together the clot—job done! Thrombin also activates the intrinsic pathway, amplifying the process. Factor XIII, produced by platelets, stabilizes the clot.

Which factor is common to both intrinsic and extrinsic pathways?

Role of Vitamin K and Calcium (LO #3)

Vitamin K. This water-soluble vitamin affects all pathways: extrinsic, intrinsic, and common. It is an important cofactor in the modification of glutamate residues (gamma-carboxylation), which enables them to bind calcium ions (Figure 5). This modification occurs in Factors II, VII, IX, and X. Warfarin is an anticoagulant medication that works by blocking an enzyme that activates vitamin K affecting both the intrinsic and extrinsic pathways.

CLINICAL CORRELATION

Warfarin is a common anticoagulant used to treat and prevent blood clots, and it works by competitively inhibiting the effects of Vitamin K (prevents activation of Vit K and depletes its functional levels). It is important for the synthesis of clotting factors and anticoagulant proteins C and S. Diet and metabolism of Vitamin K alter the amount of warfarin needed for treatment, and the warfarin dosing is critical. There are serious side effects of warfarin, where too much can result in bleeding and hemorrhage.

CLINICAL CORRELATION

Newborns lack vitamin K–producing gut bacteria at birth, predisposing them to **hemorrhagic disease of the fetus and newborn** with serious bleeding risks (especially intracranial hemorrhage). To prevent this, newborns routinely receive a vitamin K injection in the delivery room.

Calcium affects numerous parts of the clotting cascade through its role in binding to clotting factors and activating additional downstream factors. It is also released near the site of the wound. Figure 5 shows in each pathway where calcium acts. Thus, chelating calcium (e.g., with EDTA or citrate) can block all parts of the clotting cascade. This is why blood is collected in EDTA tubes, and why citrate is used in blood banking.

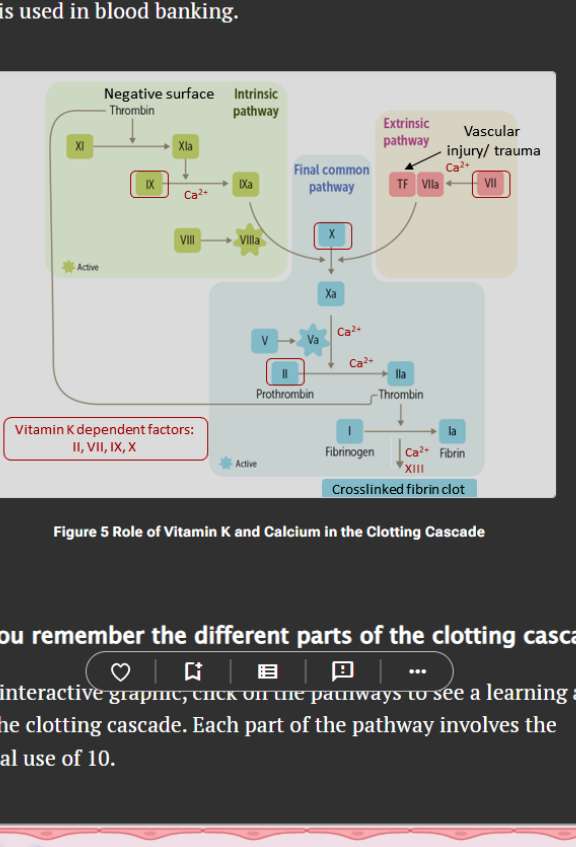
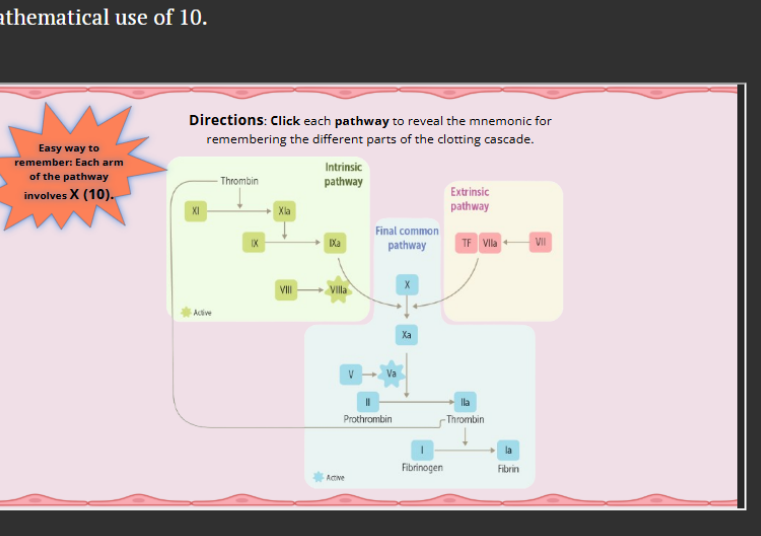


Figure 5 Role of Vitamin K and Calcium in the Clotting Cascade

How can you remember the different parts of the clotting cascade?

Below is an interactive graphic, click on the pathways to see a learning aid to remember the clotting cascade. Each part of the pathway involves the mathematical use of 10.



Easy way to remember: Each arm of the pathway involves X (10).
Directions: Click each pathway to reveal the mnemonic for remembering the different parts of the clotting cascade.
Thrombin
Intrinsic pathway
Extrinsic pathway
Final common pathway
Prothrombin
Thrombin
Fibrinogen
Fibrin
Active

How Is Blood Clotting Kept in Check? Antithrombotic mechanisms (LO #4)

The clotting cascade works when tissue and vascular injuries result in bleeding. Built in amplification mechanisms ensure that the process limits the loss of blood. Once bleeding is stopped and repair has commenced, the clots are no longer needed and the body must have a mechanism to break down and eliminate the clot. If the cascade is not kept in check under normal conditions, it can cause thrombosis, vascular inflammation, and tissue damage. There are two points where the [illegible] coagulation cascade (anticoagulants) and the clot itself (fibrinolytics).

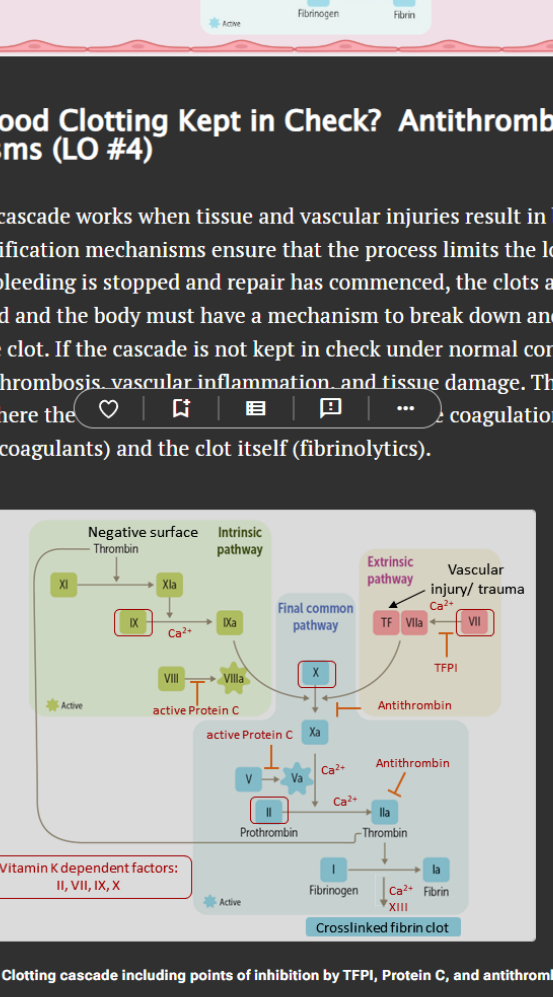


Figure 6 Clotting cascade including points of inhibition by TFPI, Protein C, and antithrombin

Preventing/Stopping the Coagulation Cascade

There are several natural anticoagulants that keep the coagulation cascade in check and stop it when it is no longer needed.

Endothelial cells have many antithrombotic effects and produce many substances that inhibit platelet binding, secretion, and activation (antiplatelet activity), such as **prostacyclin** and **nitric oxide (NO)**. They also produce anticoagulant factors such as **heparin-like molecules** (heparan sulfate). Heparin coated tubes [illegible] ting blood clotting when performing blood collection in the clinical setting.

The anticoagulants, tissue factor pathway inhibitor (TFPI), antithrombin, and protein C and S, turn off the cascade. These molecules are on standby, effectively shutting down the clotting cascade.

- **Tissue factor pathway inhibitor** (**TFPI**)—a substance produced by endothelial cells and found in plasma that switches off the extrinsic arm almost immediately after the pathway is switched on.
- **Antithrombin,** as its name indicates, is an inhibitor of thrombin. Recall that thrombin amplifies the clotting cascade. Thus, it affects both the intrinsic and final common pathways. Importantly, antithrombin activity is enhanced by the presence of heparin. Since heparin-like molecules are on the surface of endothelial cells, the activity of antithrombin is enhanced at the vessel walls.
- **Protein C** is a major regulator of the coagulation process. Protein C is a plasma glycoprotein that binds to thrombomodulin present on the endothelial cell surface. When activated by the thrombin-thrombomodulin complex, activated Protein C binds to **Protein S**, together resulting in cleavage and inactivation of Factors VIIIa and Va.

CLINICAL CORRELATION

Activated Protein C (aPC) is important in a hypercoagulable pathway called Factor V Leiden. This results in a mutation of Factor V, which is not degraded by aPC. It is the most common inherited thrombophilia (hypercoagulopathy) in Caucasian populations of European descent.

Dissolving a Clot — Fibrinolytic system

Once the vessel wall is sealed and repaired, any excess areas of clot need to be eliminated, a process called clot remodeling. Dissolving clots can also come in handy in the event that you find yourself with a pathologic clot (DVT, pulmonary embolism, arterial thrombosis).

Clots—or portions of clots—are destroyed in a process called thrombolysis or fibrinolysis. The main actor here is an enzyme called plasmin. Its precursor, plasminogen binds to fibrinogen and fibrin, and thus is incorporated into the clots as it forms. Plasmin cleaves the fibrin surrounding the clot, leaving little fragments of clot that simply wash away. Plasmin digestion of cross-linked fibrin releases D-dimers. These D-dimers are sensitive markers of blood clot formation and are used to diagnose deep vein thrombosis (DVT) and pulmonary embolism. Last, any free plasmin circulating in the blood is rapidly inhibited by alpha 2-plasmin.

Tissue plasminogen activator (tPA) is secreted from endothelial cells, and is important in converting plasminogen to active plasmin. tPA is also used clinically to dissolve clots after strokes, restoring blood flow though the blood vessel.

How is the coagulation cascade regulated?

CLINICAL CORRELATION

Virchow's Triad are clotting risk factors: **stasis** (slow or stagnant blood flow), **hypercoagulability** (↑ tendency of the blood to clot), and **endothelial injury** (damage to the inner lining of blood vessels). Oral estrogen (contraceptives, HRT) can predispose to clots (hypercoag.). Atherosclerotic plaque rupture and clotting lead to myocardial infarction (coronary arteries); & **stroke** (cerebral arteries). Clots in the legs (often due to stasis or hypercoagulability) may dislodge and travel to the lungs, causing a **pulmonary embolism.**

Coagulation tests

There are times when one would need to measure the effectiveness of the different parts of the hemostatic system (intrinsic vs extrinsic), and the clotting ability of individuals. The tests below measure the time for a clot to form. For example, people taking anticoagulant medication such as warfarin may need to have their clotting times assessed in order to adjust the dose.

- The **prothrombi**[illegible] **insic pathway** (and, by extension, the common pathway) of the coagulation cascade (Factors I, II, V, VII, and X, Figure 4). It uses a mixture of phospholipid, a **tissue factor**-like substance, and calcium. It normally takes 11-14 seconds to form fibrin. To allow for a comparison of clotting times between different labs, the **International Normalized Ratio (INR)** was developed. This is the normalization of the PT clotting time of the patient to controls.
- The **activated partial thromboplastin time (aPTT)** is used to assess the **intrinsic pathways** (and, by extension, the common pathway) of the coagulation cascade (Factors I, II, V, VIII, IX, X, XI, XII, Figure 4). The PTT uses the phospholipid agent (**negative surface**), and therefore, no activation of the extrinsic pathway occurs, as happens for the PT and INR. By standardizing the activation process, the aPTT version is more precise and reproducible and has replaced the original PTT. The normal range for the formation of a clot in the aPTT test is 25 to 35 seconds.

CLINICAL CORRELATION

An additional important test in the screening of bleeding disorders is the complete blood count (CBC), which measures the amount of hemoglobin in the red cells, the size and number of red blood cells (among other parameters), the number and types of white blood cells, and the number of platelets. A decreased platelet count (thrombocytopenia) can be associated with bleeding symptoms.

Mnemonic

What is a mnemonic to remember Warfarin, the factors it affects and clotting time measure?

Mnemonic

What is mnemonic to remember the difference between PT and aPTT clotting times?

Bleeding disorders (LO #5)

People with deficien[illegible] hilia) tend to bleed after a traumatic injury, although they may not remember the specific injury and it might have seemed insignificant. For example, they may have dental surgery and then hours later experience excessive bleeding from the surgical site. Such patients can form a platelet plug and are able to achieve temporary hemostasis right after an injury (primary hemostasis), but stabilization of the clot does not occur (secondary hemostasis).

Hereditary Bleeding disorders

von Willebrand disease is the most common hereditary bleeding disorder (occurring in about 1 in 100 people, many of whom are totally asymptomatic). It is caused by decreased functional von Willebrand factor, the protein that helps platelets bind to the sub-endothelium or a qualitative defect in the factor. Common symptoms include mucocutaneous bleeding (nose bleeds, easy bruising, petechiae, heavy menstrual bleeding, and gum bleeding). While it plays a role in platelet clotting, it also stabilizes factor VIII. Thus, the PT is normal. The PTT may be prolonged in some cases, yet it is not a reliable screening test. Testing options include vWF Antigen Test (quantification) and vWF Activity Test (function).

Hemophilia A (Factor VIII) and **Hemophilia B** (Factor IX) are inherited factor deficiencies, presenting with symptoms of muscle/joint/deep tissue bleeding. They are X-linked recessive disorders, so females are carriers and males are usually those affected. They affect the intrinsic pathway, and thus have increased partial thromboplastin time (aPTT - intrinsic pathway), with normal prothrombin (PT/INR - extrinsic pathway) bleeding time. Hemarthroses, bleeding into joints, are particularly common with the hemophilias. Figure 7 Since synovial joint spaces naturally have low levels of Tissue Factor, the extrinsic clotting pat[illegible] se with disorders of the intrinsic pathway, and joint bleeding can occur with minimal to no trauma.

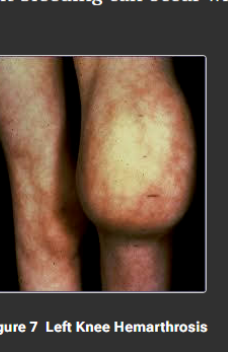


Figure 7 Left Knee Hemarthrosis

Acquired Bleeding disorders

Prolonged or severe **Vitamin K deficiency** can lead to excessive bleeding. Fat-soluble Vitamin K is essential for the synthesis of coagulation factors II, VII, IX, and X in the liver. Vitamin K is involved in modifying glutamate residues, to enhance Ca^{2+} binding. This deficiency is usually seen in extreme malnutrition, either due to insufficient intake or gut malabsorption. See the Vitamin K section above.

Liver failure (as in cirrhosis) can lead to lack of hepatic synthesis of coagulation factors made in the liver, with reduced coagulation factor levels and increased bleeding. Laboratory analysis can show elevated PT and PTT with thrombocytopenia.

Iatrogenic coagulopathy is where the coagulation pathway is inhibited by medicines such as an[illegible] ulants such as heparins and warfarin. While these are used in clinical conditions, incorrect dosage can be hazardous.

Thrombocytopenia (decreased platelet count), caused by a variety of conditions, may lead to mucocutaneous bleeding (bleeding gums, heavy menstrual periods, petechiae and purpura on the skin).

CASE CONNECTION BACK TO INTRODUCTION ↑

GC has von Willebrand's Disease, which is the most common inherited bleeding disorder, typically in an autosomal dominant pattern (it can also be autosomal recessive), unlike the hemophilias. It is frequently undiagnosed. Although the aPTT *may* be prolonged due to affects on circulating factor VIII, this test, and the PT and platelet count are usually normal.

With this family history and symptoms, further investigation is warranted prior to proceeding with an elective surgical procedure.

Confirmation typically involves Hematology consultation and special labs: vWF Antigen Test and vWF Activity Test.

Summary

- Hemostasis is the process of stopping bleeding, also known as clotting, and involves the coordinated activity of plasma components, the vascular wall, and platelets.
- The process of hemostasis includes vessel constriction, platelet plug formation, and fi[illegible] as primary and secondary hemostasis.
- Primary hemostasis involves platelets and von Willebrand factor, forming a platelet plug, while secondary hemostasis involves the formation of a fibrin clot through the activation of coagulation factors.
- The coagulation cascade is a series of plasma proteins that interact to clot the blood, starting with the tissue factor initiating the extrinsic pathway, and thrombin initiating fibrin formation along the intrinsic pathway.
- Hemostasis is regulated by natural anticoagulants like antithrombin, protein C, and protein S, and by the fibrinolytic system which dissolves clots.
- Disorders of hemostasis can be hereditary, such as von Willebrand disease and hemophilia, or acquired, such as Vitamin K deficiency, liver failure, thrombocytopenia, or due to certain medications.

Review Questions

1. Which of the following circulating molecules is responsible for dissolving an already-formed clot?

   A. ○ Antithrombin
   B. ○ Fibrinogen
   C. ● Plasmin
   D. ○ Plasminogen
   E. ○ Thrombin

Hide Explanation

The correct answer is plasmin (C). Plasmin is the main protein involved in thrombolysis, the process of dissolving clots or breaking up clots in blood vessels. Antithrombin (A) stops the formation of fibrin but does not dissolve any already-formed fibrin. Fibrinogen (B) and plasminogen (D) are inactive molecules known as zymogens and can only be activated by enzymes (many coagulation factors are enzymes). Thrombin (E) is a key factor in the coagulation cascade because it activates many other coagulation factors.

2. What role does von Willebrand factor play in primary hemostasis?

   A. ○ It activates thrombin
   B. ○ It cross-links fibrin strands
   C. ● It aids in platelet adhesion to the vessel wall
   D. ○ It dissolves fibrin clots
   E. ○ It stabilizes the platelet plug

Hide Explanation

Correct Answer C): Von Willebrand factor mediates platelet adhesion to the subendothelial matri[illegible] Von Willebrand factor does not A) directly activate thrombin. B). cross-link fibrin cross-linking (it is Factor XIII in secondary hemostasis). D). Fibrin clot dissolution is carried out by plasmin during fibrinolysis. E). Although von Willebrand factor helps initiate platelet adhesion, the stabilization of the platelet plug is primarily achieved by the formation of fibrin during secondary hemostasis.

3. Which of the following best describes the role of Vitamin K in coagulation?

   A. ○ It activates plasminogen
   B. ○ It is essential for the synthesis of fibrinogen
   C. ● It is required for the gamma-carboxylation of certain clotting factors
   D. ○ It directly activates thrombin
   E. ○ It binds to calcium ions

Hide Explanation

The correct answer is C) It is required for the gamma-carboxylation of certain clotting factors. Vitamin K is essential for modifying glutamic acid residues on factors II, VII, IX, and X, allowing them to bind calcium and function properly in the coagulation cascade. Incorrect Answers: A) Plasminogen is activated by tissue plasminogen activator (tPA) during fibrinolysis. B) Fibrinogen synthesis occurs in the liver. D) Thrombin is activated by Factor Xa in the coagulation cascade E) Vitamin K does not bind to calcium ions; it enables clotting factors to do so after gamma-carboxylatio[illegible]

4. Which of the following best explains why patients with hemophilia tend to bleed excessively?

   A. ○ They have a deficiency in platelet production
   B. ○ They are unable to initiate the extrinsic pathway
   C. ● They lack the ability to form a stable fibrin clot
   D. ○ They overproduce anticoagulants
   E. ○ They cannot activate Factor X properly

Hide Explanation

The correct answer is C) They lack the ability to form a stable fibrin clot. Hemophilia is characterized by a deficiency in clotting factors (such as Factor VIII or IX), which prevents the formation of a stable fibrin clot, leading to prolonged bleeding. Incorrect Answers: A) Hemophilia affects clotting factors. B) Hemophilia primarily affects the intrinsic pathway, not the extrinsic pathway. D) Hemophilia is not associated with overproduction of anticoagulants but with a deficiency in specific clotting factors. E) While the intrinsic pathway contributes to Factor X activation, the key issue in hemophilia is the deficiency of upstream clotting factors like Factor VIII or IX.

5. Which of the following is the first event in primary hemostasis?

   A. ○ Platelet aggregation
   B. ● Blood vessel constriction
   C. ○ Formation of fibrin
   D. ○ Activation of thrombin
   E. ○ Platelet adhesion

Hide Explanation

The correct answer is B). Vessel constriction reduces blood flow to the site of injury. Incorrect Answers: A) Platelet aggregation occurs later in primary hemostasis. C) Fibrin formation stabilization of the platelet plug in secondary hemostasis. D) Activated thrombin activation converts fibrinogen to fibrin in secondary hemostasis, E) Platelet adhesion occurs after blood vessel constriction primary hemostasis.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~List the steps involved in primary (platelet plug) and secondary (fibrin clot) hemostasis.~~
02 ☑ ~~Compare and contrast the extrinsic, intrinsic, and common pathways (clotting or coagulation cascade).~~
03 ☑ ~~Describe the importance of Vitamin K and calcium in coagulation.~~
04 ☑ ~~Describe the process and regulation of fibrinolysis and clot resolution.~~
05 ☑ ~~Discuss common disorders affecting hemostasis.~~

Objective Confidence: 5/5

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04 ☑ ~~Describe the process and regulation of fibrinolysis and clot resolution.~~
05 ☑ ~~Discuss common disorders affecting hemostasis.~~

Objective Confidence: 5/5